

cluster, I will need another two nodes to be configured as management nodes. The scenario is given below.

Management nodes

```
mgmtsrv01    172.22.10.10
mgmtsrv02    172.22.96.10
```

MySQL and data nodes

```
dhcprsv01    172.22.11.100    (Primary DHCP for Region 1)
dhcprsv02    172.22.22.100    (Primary DHCP for Region 2)
dhcprsv03    172.22.33.100    (Primary DHCP for Region 3)
dhcprsv04    172.22.44.100    (Secondary DHCP for all)
```

System: Linux based virtual machines

Operating system: CentOS release 6.7

RAM: 8GB

CPU cores: 4

Packages dependencies

1. libaio.x86_64 0:0.3.107-10.el6.rpm
2. libaio-devel.x86_64 0:0.3.107-10.el6.rpm
3. numactl-2.0.9-2.el6.x86_64.rpm
4. cryptopp-5.6.1-8.1.x86_64.rpm (required for PDNS)
5. php-pear-MDB2-Driver-mysql-1.5.0-0.8.b4.el6.noarch.rpm (required for PDNS)
6. php-pear-MDB2-2.5.0-0.7.b4.el6.remi.noarch.rpm (required for PDNS)
7. php-mysql-5.3.3-46.el6_6.x86_64 (required for PDNS)
8. php-pear-1.9.4-4.el6.noarch (required for PDNS)
9. php-pdo-5.3.3-46.el6_6.x86_64 (required for PDNS)
10. perl-DBD-MySQL-4.013-3.el6.x86_64 (required for PDNS)

Packages installed

1. MySQL-Cluster-client-gpl-7.3.11-1.el6.x86_64.rpm
2. MySQL-Cluster-server-gpl-7.3.11-1.el6.x86_64.rpm
3. MySQL-Cluster-shared-compat-gpl-7.3.11-1.el6.x86_64.rpm (required for PDNS)

Packages that need to be removed (if any)

1. mysql-server
2. mysql
3. mysql-libs

Configuration for data nodes

Edit the `/etc/my.cnf` file as shown below:

```
[mysqld]
ndbcluster
# IP address of the cluster management node
ndb-connectstring=172.22.100.10,172.22.96.10
socket=/var/lib/mysql/mysql.sock
user=mysql
innodb_data_file_path = ibdata1:10M:autoextend
pid-file=/var/run/mysqld/mysqld.pid
explicit_defaults_for_timestamp = 1

[mysql_cluster]
```

```
# IP address of the cluster management node
ndb-connectstring=172.22.100.10,172.22.96.10
```

```
[mysqld_safe]
log-error=/var/log/mysqld.log
pid-file=/var/run/mysqld/mysqld.pid
```

Configuration for management nodes

Edit the `/var/lib/mysql-cluster/config.ini` file as shown below:

```
[ndbd default]
NoOfReplicas=4
DataDir=/opt/mysql-cluster/data
MaxNoOfAttributes=10000
MaxNoOfConcurrentOperations=150000
MaxNoOfOrderedIndexes=512
```

```
# MANAGEMENT NODES
[ndb_mgmd]
NodeId=1
hostname=172.22.100.10
datadir= /var/lib/mysql-cluster
```

```
[ndb_mgmd]
NodeId=2
hostname=172.22.96.10
datadir= /var/lib/mysql-cluster
```

DATA NODES

```
[ndbd]
NodeId=3
hostname=172.22.11.100
datadir=/usr/local/mysql/data
```

```
[ndbd]
NodeId=4
hostname=172.22.44.100
datadir=/usr/local/mysql/data
```

```
[ndbd]
NodeId=5
hostname=172.22.33.100
datadir=/usr/local/mysql/data
```

```
[ndbd]
NodeId=6
hostname=172.22.100
datadir=/usr/local/mysql/data
```

```
#MYSQL NODES
[mysqld]
NodeId=7
hostname=172.22.11.100
```